TNT Cheap EEG Kahoot: answer key

Papers: Krigolson et al. 2017 (Muse), Frey 2016 (OpenBCI vs g.tec), Pontifex & Coffman 2023 (Unicorn), Chevallier et al. 2024 (MOABB benchmark); plus the EEG artifact video.

1. Krigolson's Muse cost about $249. The lab EEG it was compared to cost about...
- $2,000
- $10,000
- **$77,000 (correct)**
- $500

Why: About $77,100 for the lab's ActiChamp system: roughly 300 times the price.

2. Which brain responses could the 5-channel Muse detect?
- **N200, P300 and reward positivity (correct)**
- None at all
- Only alpha waves
- Only SSVEP

Why: All three ERPs showed up, from about 10 participants.

3. Compared with the lab system, the Muse's ERPs were...
- Exactly the same
- Larger
- **Smaller (correct)**
- Impossible to see

Why: Smaller effects (P300 about 1.4 vs 10.5 microvolts), partly because Muse has no Pz electrode.

4. How did Frey compare OpenBCI and a $20k g.tec amp fairly?
- Different people on each
- **Same electrodes, recorded at the same time (correct)**
- Different days
- Simulated data

Why: One set of electrodes was split to feed both amplifiers at once.

5. P300 decoding (AUC): g.tec got 0.961. What did the ~800 euro OpenBCI get?
- 0.50
- 0.70
- **0.918 (correct)**
- 0.999

Why: Close to lab-grade, though still a bit lower.

6. What is the biggest weakness of Frey's study?
- **Only one participant (correct)**
- Too many channels
- No P300 task
- Used dry electrodes

Why: One person, one session: a promising result, not a final answer.

7. Pontifex: what made the Unicorn's P300 data valid?
- Dry electrodes
- More channels
- Wi-Fi streaming
- **Salt water on the electrodes (correct)**

Why: Saline raised the ERP correlation with the lab system from .55 to .78.

8. To match the lab system's P3 power, the Unicorn needed about how many times more trials?

The same
About 8x (correct)
About 2x
About 100x
Why: Cheap hardware can get there; it just costs more data.

9. Which measures held up even with DRY Unicorn electrodes?
Spectral measures like alpha (correct)
ERPs like the P300
Nothing
Only EMG
Why: Spectral correlation with the lab system was .84 dry, .90 with saline.

10. MOABB benchmark, 36 datasets: which model family did best overall?
Deep learning
Random forests
Riemannian (correct)
Guessing
Why: Covariance + tangent space pipelines ranked highest across MI, P300 and SSVEP.

11. Why did deep learning lag in the MOABB benchmark?
Python is too slow
Too few trials per class (correct)
Too many electrodes
Wrong paradigm
Why: Deep models need more data, often 150+ trials per class, and tuning.

12. In the artifact video, a jaw clench mostly shows up as...
A slow bump below 5 Hz
Alpha at 10 Hz
Nothing at all
Fast spiky activity, 20+ Hz (correct)
Why: Muscle activity is fast and broadband; blinks are the slow bumps.

13. Across these papers, cheap EEG works best on...
Strong signals: alpha, P300, SSVEP (correct)
Tiny subtle effects
Reading words from thoughts
Nothing useful
Why: That's our hypothesis to test: which paradigms survive cheap hardware?